

Subject Analysis - Room Booking Database

1. Candidate Subjects

Entities and Key Attributes

Building

Attribute	Type	Notes
building_id	PK	Primary key
name	NOT NULL	Building name
address	NOT NULL	Physical address

Room

Attribute	Type	Notes
room_id	PK	Primary key
building_id	FK	-> Building
name_or_number	NOT NULL	Room identifier within the building
capacity	NOT NULL	Maximum number of people
type	NOT NULL	meeting room / lecture room / studio / workshop room
requires_approval	NOT NULL	Boolean - whether this room requires special access approval

EquipmentType

Attribute	Type	Notes
equipment_id	PK	Primary key
name	NOT NULL	e.g. projector, whiteboard, microphone, video camera

RoomEquipment (linking table)

Attribute	Type	Notes
room_id	FK	-> Room
equipment_id	FK	-> EquipmentType

Organization

Attribute	Type	Notes
organization_id	PK	Primary key
name	NOT NULL	Student organization name

Person

Attribute	Type	Notes
person_id	PK	Primary key
name	NOT NULL	Full name
email	NOT NULL	Contact email - minimum personal data only

BookingSeries (the planning intent for recurring bookings)

Attribute	Type	Notes
series_id	PK	Primary key
room_id	FK	-> Room
organization_id	FK, nullable	-> Organization; NULL if series is created by an individual with no organizational affiliation
created_by	FK	-> Person
recurrence_rule	NOT NULL	e.g. "weekly", "biweekly"
series_start_date	NOT NULL	First occurrence date
series_end_date	NOT NULL	Last occurrence date
day_of_week	NOT NULL	e.g. "Wednesday"
start_time	NOT NULL	Time of day the slot starts
end_time	NOT NULL	Time of day the slot ends

Why this table exists: The case explicitly requires that a recurring booking be treatable *"both as a single planning intent and as a series of individual occurrences."* BookingSeries stores that intent. Without it, there is no single fact in the database representing "the organization has booked Wednesday evenings." The pattern fields (recurrence_rule, day_of_week, start_time, end_time) belong here because they describe the original intent, not any individual occurrence.

Booking (one time slot - standalone or one occurrence of a series)

Attribute	Type	Notes
booking_id	PK	Primary key
room_id	FK	-> Room

Attribute	Type	Notes
organization_id	FK, nullable	-> Organization; NULL if booking is made by an individual with no organizational affiliation
created_by	FK	-> Person
start_datetime	NOT NULL	Start of the booking slot
end_datetime	NOT NULL	End of the booking slot
status	NOT NULL	pending / confirmed / cancelled / rejected
series_id	FK, nullable	-> BookingSeries; NULL if standalone booking
created_at	NOT NULL	Timestamp when the booking was created - needed for reporting and auditing
cancelled_at	nullable	Timestamp of cancellation; NULL if not cancelled
cancellation_reason	nullable	Short free text; NULL if not cancelled
requires_approval	NOT NULL	Snapshot of Room.requires_approval at booking creation time
approval_granted	nullable	NULL if approval not required or not yet decided; TRUE if approved; FALSE if rejected

Why requires_approval appears on both Room and Booking: Room.requires_approval reflects the room's current policy. Booking.requires_approval is a snapshot taken at booking creation time. If the room's policy changes later, historical bookings are not affected. This is a justified denormalization, noted in the design-quality review.

Reading approval state:

requires_approval	approval_granted	Meaning
FALSE	NULL	No approval needed - booking may be pending or confirmed
TRUE	NULL	Pending - waiting for approval decision
TRUE	TRUE	Approved
TRUE	FALSE	Rejected

2. Candidate Relationships

From	Relationship	To	Cardinality	Notes
Building	contains	Room	1 to many	A room belongs to exactly one building

From	Relationship	To	Cardinality	Notes
Room	has	EquipmentType	many to many	Via RoomEquipment linking table
Person	creates	Booking	1 to many	Minimal personal data stored
Person	creates	BookingSeries	1 to many	Same person model used for series creation
Organization	associated with	Booking	0 or 1 to many	Organization linked via FK; nullable - a booking may have no organization if made by an individual
Organization	associated with	BookingSeries	0 or 1 to many	Organization linked via FK; nullable - a series may have no organization if created by an individual
Room	is booked by	Booking	1 to many	A booking covers exactly one room
Room	is target of	BookingSeries	1 to many	A series targets one room
BookingSeries	consists of	Booking	1 to many	A standalone booking has series_id = NULL

3. Key Assumptions

1. **Equipment is modeled as types, not physical assets.** A room either has a projector or does not. Serial numbers and individual item tracking are out of scope.
2. **A standalone booking and a recurring occurrence share the same Booking table.** A standalone booking simply has series_id = NULL.
3. **requires_approval is snapshotted onto Booking at creation time.** This ensures historical bookings are not affected if a room's approval policy changes later.
4. **approval_granted on Booking covers both the "required" and "outcome" states** via a nullable boolean. No separate approval table, no approver identity, no workflow stages - the case only asks whether approval was required and whether it was granted.
5. **Organization is a separate entity**, not just a name string on the booking. This makes reporting on organizations clean and consistent.
6. **organization_id is nullable on both Booking and BookingSeries.** A booking or a recurring series may be made by an individual with no organizational affiliation. This applies equally to standalone and recurring bookings - a person should be able to book a room for themselves in both cases.

7. **Person stores name and email only.** This is the minimum needed to manage bookings responsibly, as stated in the case.
8. **Double-booking is prevented** - a room cannot have two confirmed bookings with overlapping time slots. This is enforced at the schema or application level.
9. **Recurrence is stored as a descriptive rule string and date range** (e.g. "weekly", start date, end date, day of week). The database does not parse or enforce the rule - it is descriptive only.
10. **created_at is recorded on every Booking** to support reporting (e.g. bookings made per month) and auditing.

4. Ambiguities and Chosen Interpretations

Ambiguity	Chosen Interpretation
Is equipment a specific physical item or a type?	Equipment type per room. No serial numbers or individual asset tracking.
Does a recurring booking need its own table?	Yes - BookingSeries stores the planning intent; individual Booking rows are the occurrences. Required explicitly by the case description.
What personal data is stored about the person creating a booking?	Name and contact email only.
Can the same room be double-booked?	No. Enforced as a constraint on confirmed bookings.
What does "approval" mean in practice - workflow, approver identity?	We store only requires_approval (snapshot boolean) and approval_granted (nullable boolean). No approver identity or multi-step workflow.
Can a cancellation reason contain personal information?	Possibly. We store it as a short optional free-text field and flag it as a privacy risk in the design-quality review.
Is recurrence pattern always weekly, or flexible?	We store a recurrence_rule string (e.g. "weekly", "biweekly") and a date range. The design does not enforce or parse the rule.
Is organization a separate entity or just a name string?	Separate entity with its own table, to support clean reporting by organization.
Should requires_approval live only on Room or also on Booking?	Both - Room.requires_approval is the current policy; Booking.requires_approval is a snapshot at creation time. Justified denormalization.
Can a booking be made without an organization?	Yes for both standalone bookings and recurring series - organization_id is nullable on both Booking and BookingSeries. A person may book a room for themselves without any organizational affiliation.

5. Likely Application Queries

These are the queries the schema must be able to answer. All must be traceable through the conceptual model.

ID	Query	Source
Q1	How many bookings has each room had?	Case description
Q2	Which organizations use the service most?	Case description
Q3	Which rooms are most heavily used (by total hours booked)?	Case description
Q4	Which equipment types appear most often in booked rooms?	Case description
Q5	How many bookings were cancelled?	Case description
Q6	Which bookings required approval, and was it granted?	Case description
Q7	List all occurrences of a given booking series	Derived from recurring booking requirement
Q8	Which bookings were made by a given organization within a date range?	Derived from reporting requirement